RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI-602105

CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

SUBMITTED BY

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BACHELOR OF ENGINEERING

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INVESTIGATION OF POSSIBLE UNAUTHORIZED ACCESS TO A COLLEGE SERVER

UNIQUE CASE SCENARIO

Scenario: A possible late-night transfer of examination-paper drafts. A departmental server records unusual access to the confidential “ExamDrafts” folder. The investigation considers an authorized staff workstation and a recovered USB drive to assess whether files may have been copied.

SCOPE AND ASSUMPTIONS

The administrator has authorized an investigation limited to the relevant systems and time period. The sample logs and filenames are illustrative; they do not establish that a real person copied files or was responsible.

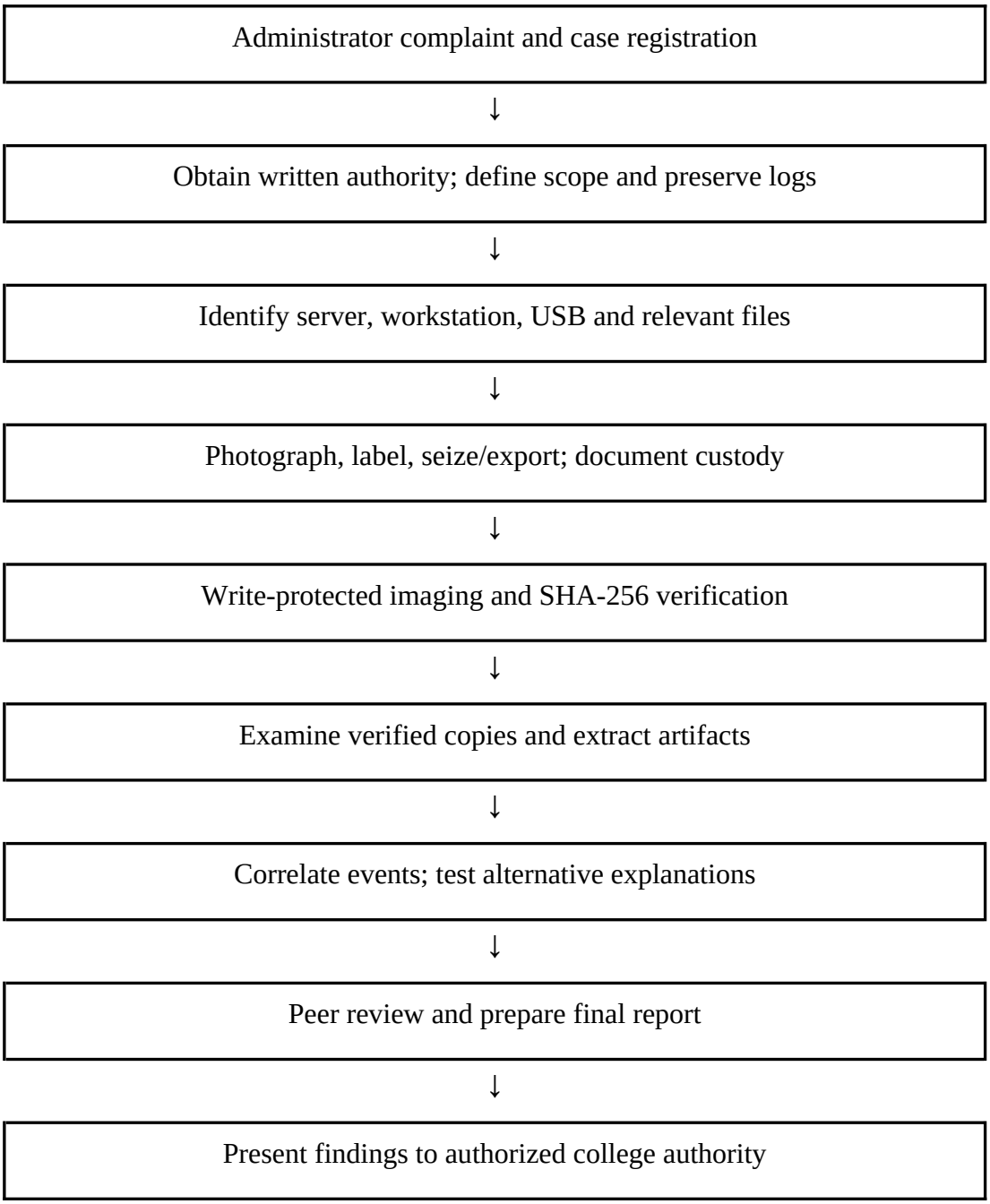
Question 1. Describe the Identification, Collection, Examination, Analysis and Presentation stages.

Identification: register the complaint, define the suspected time window, identify the server, relevant accounts, workstation, USB, file shares, logs and backup sources; obtain authority and preserve short-lived logs. Collection: photograph and inventory devices, capture volatile information only when justified, export server/security logs, acquire forensic images with write protection, calculate SHA-256 hashes and maintain custody records.

Examination: work from verified copies; extract file-system metadata, authentication records, USB connection traces, shortcuts and file hashes without altering originals. Analysis: correlate login sessions, share access, file reads, possible USB writes and account ownership; test innocent explanations such as scheduled backups. Presentation: provide a reproducible evidence list, timeline, methods, limitations and a plain-language finding distinguishing access from proven copying.

Question 2. Draw a flowchart for the forensic investigation.

The flowchart below presents the suggested investigation sequence. Authorization and evidence-integrity checks must be completed at the relevant stages.



Question 3. List and classify five or more possible evidence items.

Each item is classified according to the investigative question it may help answer. Its presence alone does not establish wrongdoing.

Item	Investigative value	Interpretation / limitation
Administrator complaint	Lead / contextual	Defines allegation and time range; not proof

Server access/security logs	Primary activity record	May identify account, host and file operation; logging coverage varies
Workstation forensic image	Corroborating / attribution	User profiles, recent files, USB traces and local copies
Recovered USB device	Potential transfer evidence	Matching file contents and metadata can support copying
Confidential exam files	Reference / comparison	Cryptographic hashes allow exact content comparison
Authentication / network logs	Corroborating	Map session and source IP; IP does not establish person
Chain-of-custody forms	Integrity / procedural	Show possession and transfers, not underlying guilt

Question 4. Explain collection and preservation precautions.

Secure the scene and document powered-on state before deciding whether to isolate or shut down; sudden shutdown can destroy volatile data, while leaving a device running can change evidence. Use approved acquisition methods and tested hardware/software write blockers for offline media; do not open original files in ordinary applications.

Photograph connections, record serial numbers, bag and seal each device separately, protect from electrostatic damage and extreme conditions, restrict access, maintain synchronized timestamps and custody logs, hash each image and verify it after copying and transfer. Record tool versions, settings, errors, bad sectors and any unavoidable alteration. Keep originals sealed and analyze read-only working copies.

Question 5. Describe a timeline of suspected access.

Normalize timestamps to a documented reference such as UTC, retaining source timezone, clock offset and timestamp semantics. Combine server logon events, share/file audit entries, workstation logons, recent-file traces, USB attachment records and USB file metadata.

Example hypothesis: 22:08 account authenticates → 22:11 confidential directory opened → 22:13 exam file read → 22:16 USB attached → 22:18 matching file appears on USB → 22:22 device

removed. These times are illustrative and must not be asserted as facts; creation times can be inherited and access times may be disabled. Seek independent corroboration and flag gaps.

Question 6. Explain the final report and presentation.

1. case identifier, investigator and authorization
2. allegation, scope and assumptions
3. exhibit register and custody history
4. acquisition tools, versions, hashes and verification
5. examination steps and reproducible artifact references
6. normalized timeline
7. findings with confidence and competing explanations
8. limitations, preservation recommendations and appendices.

Explain to the college authority what the records demonstrate (for example, “the account accessed the directory”) versus what they cannot demonstrate (“who was at the keyboard”). Use a simple timeline and a short evidence-to-conclusion table rather than unsupported technical claims.

Question 7. Discuss legal and ethical considerations.

Obtain institutional approval and follow applicable law, college policy and due-process requirements. Limit collection to systems, accounts, dates and data within the authorization; protect student examination information and third-party personal records.

Document necessity and proportionality, restrict access on a need-to-know basis, encrypt stored evidence and observe retention rules. Avoid covertly accessing unrelated private accounts or bypassing credentials without proper authority. Preserve exculpatory as well as incriminating material, disclose uncertainty, and ensure fair treatment of the employee or student under investigation.

CASE ANALYSIS AND SAMPLE REPORT SUMMARY

Potential finding: a verified server read, a temporally corresponding workstation session and a hash-matching USB copy would collectively support an unauthorized-transfer hypothesis.

Without those corroborated artifacts, the appropriate conclusion is “suspected access under investigation,” not confirmed theft. Sample report fields: Case ID CDF-01; allegation; exhibits

E01–E06; acquisition/hash register; timeline; findings; limitations; recommendations; examiner signature and date.

LINKING A SUSPICIOUS FILE TO A USER

UNIQUE CASE SCENARIO

Scenario: The shared-office desktop executable. A shared Windows desktop contains “sync_helper.exe”, hidden working files and encrypted archives; investigators test whether activity belongs to a particular account or to another user of the shared machine.

SCOPE AND ASSUMPTIONS

User identifiers, paths and times are illustrative. ILookIX is discussed as an intended examination workflow; availability and feature names should be checked against the licensed version used in the laboratory.

Question 8. Document the hardware and software components.

Record manufacturer, model, asset tag, serial number, motherboard/firmware version, clock and timezone, RAM, storage device type/capacity/serial number, partitions, attached media and network interfaces.

Record operating-system edition/build, installed applications, security tools, local/domain accounts, file-system type, encryption status, browser versions and recent updates. Photograph the workstation and capture a verified system inventory from the forensic image; do not install inventory software on the suspect disk.

Component	Illustrative record	Why relevant
Storage	SSD 512 GB; serial to be recorded	Source identity and partition layout
OS	Windows 11; exact build to be verified	Artifact paths and event interpretation
File system	NTFS; volume ID to be recorded	MFT, timestamps, ACLs and USN journal
Accounts	Staff-A, Staff-B, shared-admin	Attribution and permissions

Applications	Browser, archive tool, security agent	Execution and file-handling traces

Question 9. Explain how hardware, OS, file systems and metadata locate evidence.

Physical disk and partition maps identify relevant volumes and possible unallocated space. The operating system determines where user profiles, registry hives, event logs and application traces reside.

On NTFS, examine $MFT entries, directory paths, attributes, access-control lists, alternate data streams, $LogFile and $UsnJrnl when present. Correlate file creation/modification timestamps, owner/security identifiers, link files, jump lists, Prefetch where enabled, browser downloads and logon records. Metadata is suggestive, not automatically proof of who created a file; copying and system operations may modify it.

Question 10. Discuss hidden and encrypted files and handling.

Hidden files may be concealed by attributes, obscure paths or alternate data streams; acquire the entire volume and inspect file-system records rather than relying on Explorer visibility. Encrypted files may be inaccessible without keys, credentials or a live-session key capture lawfully authorized at collection.

Preserve ciphertext exactly, note encryption format and header information, and seek authorized recovery keys or backups. Never brute-force or modify originals without explicit scope approval; hash exported items and distinguish inaccessible content from absent evidence.

Question 11. Prepare an evidence-linkage matrix.

Each hypothetical artifact contributes a separate, testable connection; several independent sources are needed for meaningful attribution.

Artifact	Example path / time	Account linkage	Strength / caveat
Executable	C:\Users\Staff-A\Downloads\sync_helper.exe; 09:12	Profile path + browser download	Moderate; others may use profile

Prefetch / execution trace	SYNC_HELPER.EXE-*.pf; 09:17	Possible local execution	Strong for execution, weak for human identity
Hidden configuration	C:\Users\Staff-A\AppData\...; 09:18	Profile location / owner SID	Moderate; malware can write files
Encrypted archive	D:\Shared\exam_backup.7z; 09:24	Recent-file shortcut in Staff-A profile	Moderate; shortcut alone not proof of content
Logon event	Security log; Staff-A session 09:05	Session context	Supporting; session could be unattended
Browser history	Download URL at 09:12	Correlates acquisition	Supporting; browser sync may affect origin

Question 12. Explain a suitable ILookIX workflow.

1. Obtain authority and verify the forensic image hash.

2. Start a case, register exhibit identifiers and import the image through the supported evidence-import interface.

3. Confirm partition/file-system parsing and review user profiles and directory structures.

4. Filter for executables, hidden attributes, encrypted archives and target paths.

5. Inspect metadata, recoverable deleted entries, keyword hits and supported activity artifacts.

6. Bookmark relevant findings and export file listings/reports with paths, times, hashes and examiner notes.

7. Independently validate important results with another parser or raw file-system examination. Exact menus and supported functions depend on the ILookIX release; no operation is represented as performed.

Question 13. Differentiate direct activity links from contextual artifacts.

A direct activity link records the action at issue with relatively specific identifiers, such as an authenticated application audit event showing an account downloaded a named file, or a validated file operation under a session.

Contextual evidence, such as a file existing in a profile, a logged-in account or a browser visit, supports a theory but does not independently establish the action or person. Even a direct account-to-action record is not necessarily direct person-to-action evidence; credentials can be shared or stolen.

Question 14. Identify limitations and alternative explanations.

A shared desktop or unattended session may allow another person to use the account; malware or scheduled software may create files automatically; browser synchronization, copying or restore operations can alter timestamps; system clocks may drift; and logs can be incomplete or overwritten.

Encryption prevents content confirmation without keys. Attribution should therefore be expressed as a confidence-based inference supported by multiple independent artifacts, not as a certainty based on a username or path alone.

CONCLUSION WITH LIMITATIONS

The strongest defensible conclusion would be that the executable was present and possibly executed within a particular account session, if validated logs and execution artifacts agree. Establishing that the account holder personally operated it requires additional evidence, such as physical access records or reliable contemporaneous activity records.

FORENSIC IMAGING AND SAFEKEEPING

UNIQUE CASE SCENARIO

Scenario: The seized research laptop and attached USB. An investigator secures a powered-off laptop and USB drive after a report of unauthorized transfer of restricted research documents.

SCOPE AND ASSUMPTIONS

All identifiers and custody times below are illustrative. IXImager and ASB are syllabus-specific references; exact implementation details and supported options require confirmation from tool documentation and laboratory configuration.

Question 15. Step-by-step collection and transportation procedure.

1. Confirm legal authority, scene safety and collection scope.

2. Photograph the laptop, display, ports, cables and USB position; record time, power state and visible activity.

3. If powered on, decide with a trained examiner whether volatile data or encryption keys must be preserved before shutdown; document each action.

4. Isolate network connections in a manner consistent with the approved plan.

5. Label laptop, power supply and USB as separate exhibits; record serials and condition.

6. Use antistatic protective packaging, tamper-evident seals and padded transport.

7. Complete collection/custody forms and transfer promptly to secure, access-controlled storage.

8. Inspect seal integrity before laboratory acquisition.

Question 16. Explain physical acquisition and analysis of images.

Physical acquisition captures accessible sectors of a storage device, including allocated and potentially unallocated areas, rather than selecting only visible user files. This supports deleted-file recovery, partition examination and independent reanalysis.

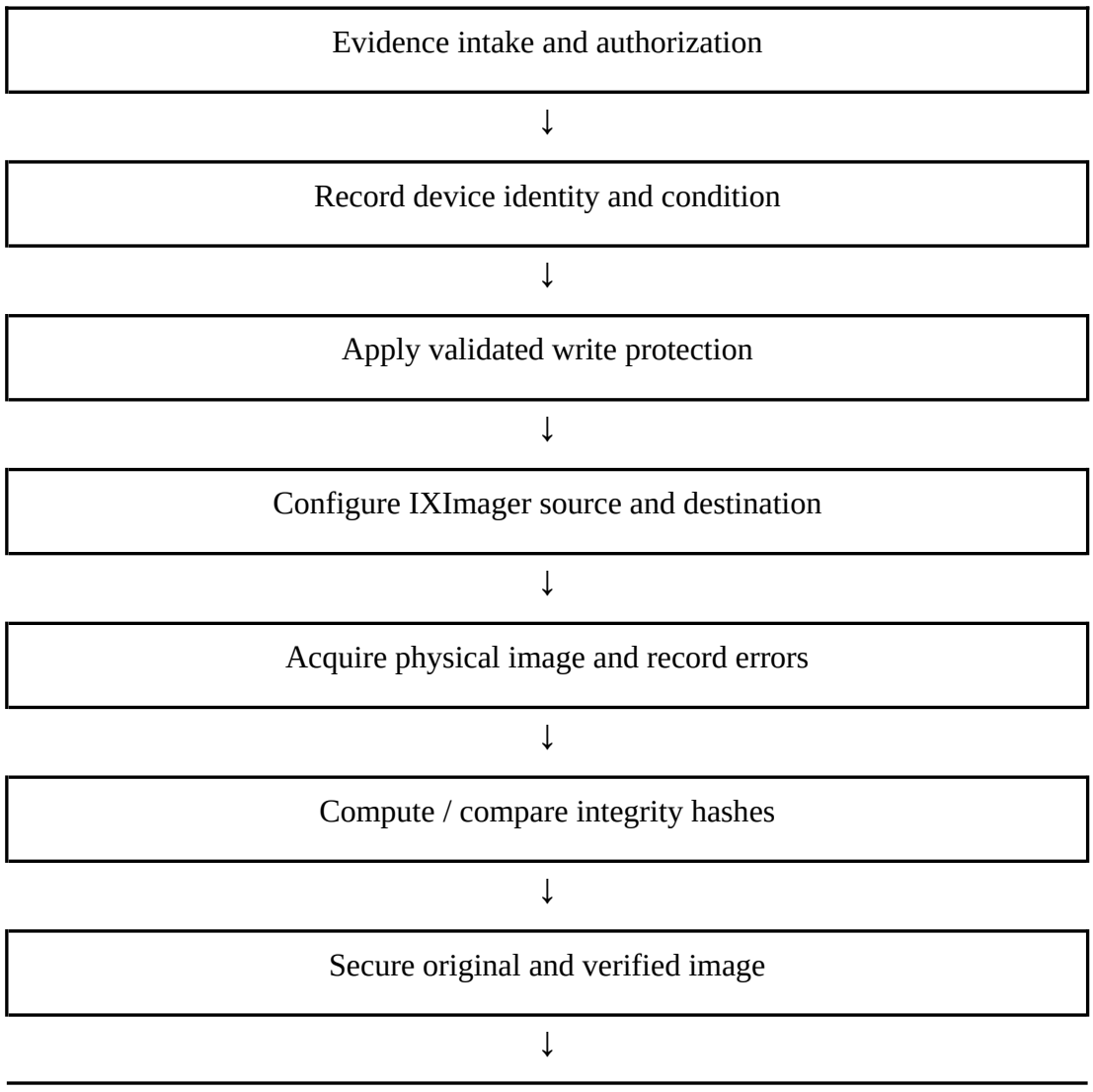
A forensic image protects the original from ordinary read/write effects and allows repeatable analysis and verification. Hardware failures, encryption and device firmware can limit readable content; record unreadable sectors and do not assume an image necessarily captures every inaccessible physical cell.

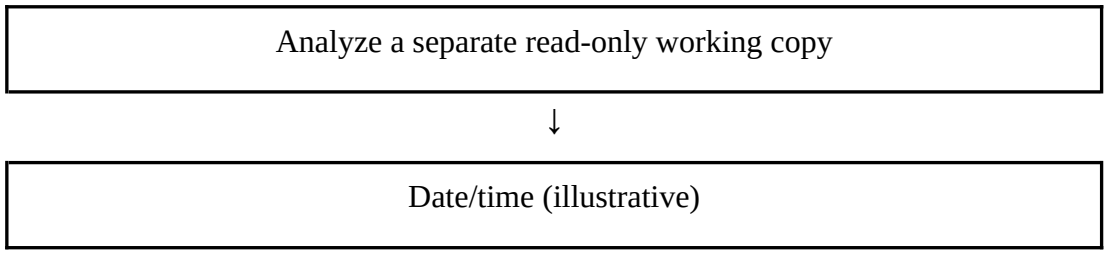
Question 17. Design an IXImager imaging workflow.

Identify the source by serial, capacity and interface; connect through a tested write blocker where feasible; record blocker self-test and source read-only status. In IXImager, configure the supported physical-source acquisition, case/exhibit metadata, destination path, image format and verification options. Confirm free destination space, start acquisition, capture logs and note errors or bad sectors.

Calculate a SHA-256 hash of source-readable data and verify the acquired image using the tool's supported verification method; document precisely what was hashed and whether hashes are comparable. Export the acquisition log, seal originals and mount verified copies read-only for examination. If ASB output is supported, document its packaging and verification behavior rather than assuming undocumented capabilities.

ACQUISITION FLOWCHART





Question 18. Explain hash-based integrity.

A cryptographic digest such as SHA-256 serves as a reproducible fingerprint of an exact byte sequence. If the acquisition tool computes comparable source and image hashes and they match, this supports accurate copying of the readable source content.

Recalculate the image hash after transfer and before examination to detect changes. Store digest, algorithm, time, tool version, input scope and examiner identity. Hash equality does not establish lawful collection, ownership or that the source was unchanged before seizure; bad sectors, container headers and segmented images require carefully defined hash scopes.

Question 19. Explain ASB container role and accompanying information.

An ASB evidence container, when supported by the selected forensic software, can be used to package/manage acquired evidence and associated case information.

Investigators must verify its exact specification, compatibility and integrity functions before relying on it; do not assume ASB has the same features as E01 or AFF. Accompany the stored evidence with case/exhibit ID, source serial and size, acquisition date/time/timezone, examiner, tool/version, source and output hashes with scope, write-blocker details, error log, storage location, custody history, access controls and any encryption credentials stored separately.

Question 20. Explain data carving and when it is needed.

Data carving searches raw byte ranges for recognizable file signatures, structures or content when directory entries and file-system metadata are deleted, corrupted or unavailable. It may help recover images, documents or archives from unallocated space or damaged partitions.

A carved item may lack its original name, path, timestamps or complete content; fragmented files can be reconstructed incorrectly. Validate signatures, internal structure and content, record offsets and avoid presenting a carved file as proof of original ownership without corroboration.

Question 21. Discuss date/time problems and documentation.

Sources may record UTC, local time, daylight-saving adjustments or device-specific epochs; clock drift, wrong time settings, timestamp resolution, filesystem semantics and synchronization delays can produce apparent contradictions.

Preserve raw timestamps and metadata, record the device timezone and observed clock offset at collection when safely measurable, document conversion formulas and uncertainties, and avoid silently normalizing ambiguous values. Distinguish file creation, modification, metadata change and access times; an acquisition or restore operation can affect them.

Question 22. Prepare a chain-of-custody table.

A chain of custody is a continuous record of who possessed the laptop, when and where each transfer occurred, why it was transferred, and whether the evidence seal remained intact. The following entries form a hypothetical example, not a record of an actual seizure. Every handover must be signed by both the releasing and receiving personnel.

At collection, assign the laptop exhibit ID LAP-01, record the make, model, serial number, power state, date, time, location and seal ID, and photograph its condition. During transport, use tamper-evident packaging and record the courier and receiving officer. At the laboratory, inspect and log the seal before controlled opening; document any seal replacement and preserve the original packaging.

During imaging, record the examiner, source identifier, write-blocking method, acquisition start/end times, imaging tool and version, output image ID and SHA-256 verification results. After examination, reseal the original, update the evidence register and return it to secured storage. Any discrepancy, broken seal or undocumented gap must be explicitly reported rather than concealed.

EVIDENCE-INTEGRITY EXPLANATION

An unbroken custody trail plus validated write protection, acquisition logs and matching hashes supports reliability. Each mechanism addresses a different risk; none alone proves the truth of an allegation.

RECOVERY FROM VOLUME SHADOW COPIES

UNIQUE CASE SCENARIO

Scenario: The overwritten procurement proposal. A Windows employee profile contains indications that a proposal was edited and removed after an unapproved cloud upload; earlier versions may survive in a Volume Shadow Copy.

SCOPE AND ASSUMPTIONS

The file operations and times are hypothetical. Shadow copies are point-in-time snapshots, not guaranteed backups, and their availability depends on system configuration and retention.

Question 23. Explain evidence selection and categorization.

Begin with the allegation and a written relevance test: whether an artifact can establish document identity, user action, deletion, cloud transfer or chronology.

Preserve the complete image first, then create an analysis index of likely paths, user profiles, event logs, browser data and shadow-copy snapshots. Mark relevance, sensitivity, source, time range and confidence. “Removing irrelevant data” should mean filtering it from the working view or report, not deleting it from the preserved image; retain auditability and protect unrelated personal information.

Question 24. Prepare a classification scheme.

Classify artifacts by function and investigative question; a single item may belong to more than one analytical category.

Class	Example	Investigative use
User-created	Proposal.docx, spreadsheets	Document content and authorship clues
System / application	Windows event logs, browser databases	Execution and session context
Potentially suspicious	Unknown upload script, archive	Possible preparation or transfer

Deleted / recovered	Deleted MFT entry, shadow-copy version	Historical state and recovery
Supporting / contextual	USB history, shortcuts, cloud logs	Cross-check activity and timeline

Question 25. Explain Event Analysis.

Event Analysis reconstructs ordered activity from heterogeneous records: logon events, file-system changes, browser downloads/uploads, application logs, shadow-copy creation and deletion indicators.

Extract source timestamps and meanings, normalize with explicit offsets, identify gaps, and group events by account, device and file identity. Test causal hypotheses rather than assuming proximity proves causation; a snapshot records state at its creation time, not necessarily the exact time a file was deleted.

Question 26. Explain Cloud Analysis and Lead Analysis.

Cloud Analysis examines authorized cloud-provider audit records, account activity, synchronization events, sharing permissions, object versions, IP addresses and access logs to ask whether a document was uploaded, shared, accessed or deleted remotely. Lead Analysis organizes promising investigative clues—such as an unusual filename, account, destination domain or time window—into prioritized follow-up actions.

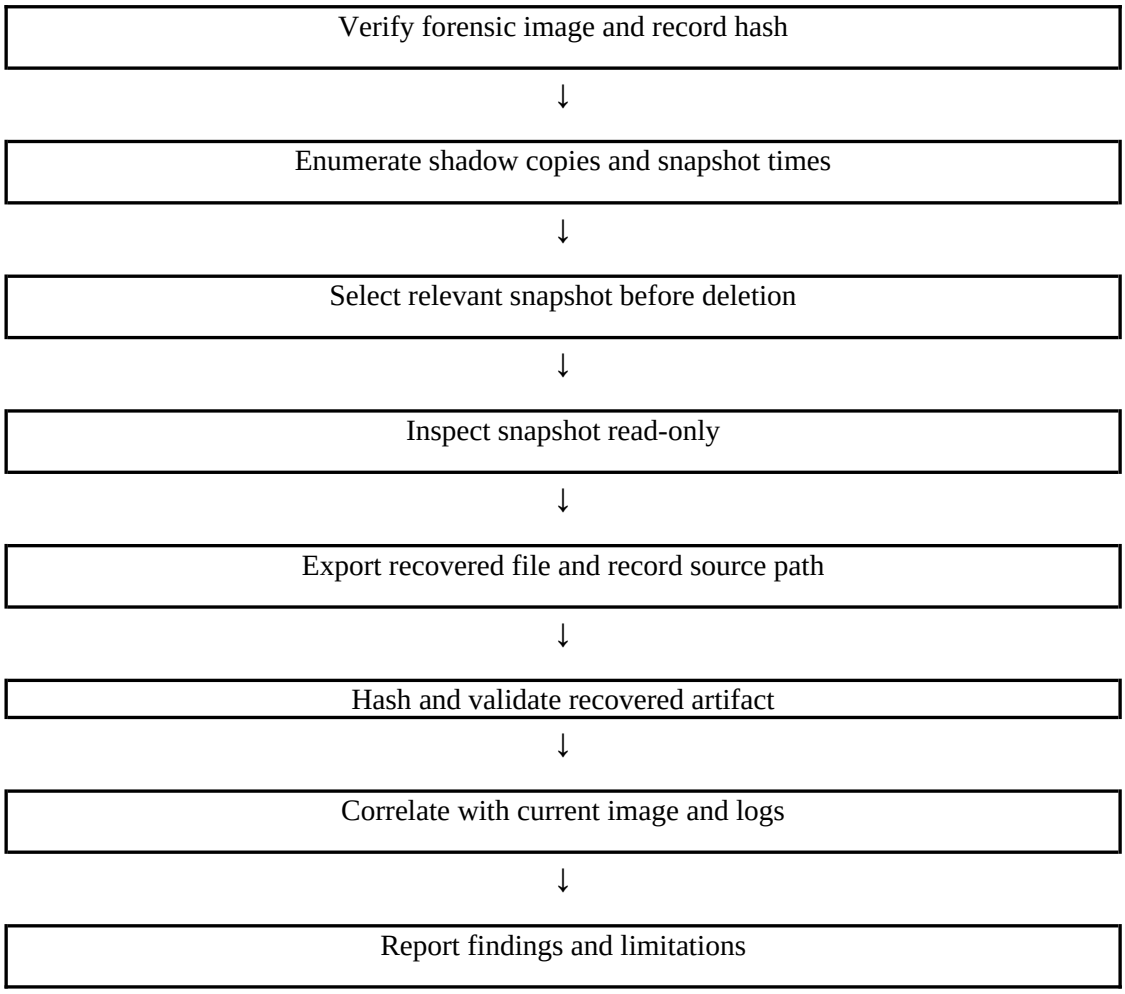
A lead guides verification; it is not itself a finding. Both processes must account for log retention, authorization and inconsistent identifiers.

Question 27. Describe Volume Shadow Copy recovery.

Verify the Windows image, enumerate available Volume Shadow Copy snapshots and their creation times using a suitable read-only forensic parser. Inspect the earlier snapshot’s file-system view for the target path; compare it with the current volume, export candidate versions and hash each export.

Compare content, size, metadata and snapshot timestamps, then correlate with event logs and cloud records. A snapshot can show that a version existed when captured, but by itself may not establish who later deleted or transferred it.

VOLUME SHADOW COPY RECOVERY WORKFLOW



Question 28. Construct a hypothetical timeline.

All times below are illustrative local times on a single day; the examiner must retain the original timezone and offset.

Time	Hypothetical event	Possible corroboration
09:00	Proposal.docx created	MFT record and application recent files
09:30	Document modified	Content and modified timestamp
10:00	Shadow copy created	Snapshot metadata; historical file version

10:20	Document copied to staging folder	USN journal / file traces if retained
10:35	Local working copy deleted	Deletion entry or missing current file
14:00	Examiner recovers 10:00 version	Snapshot export, hash and provenance

Question 29. Explain validation of recovered artifacts.

Checklist: verify parent image hash; record snapshot identifier and creation time; record original path and export method; compute exported file SHA-256; verify file signature, internal format and readability; compare with other versions or known references; inspect metadata consistency; independently corroborate significant timeline claims; record tool/version and examiner notes; document incomplete content or uncertain provenance. Preserve the unmodified export and cite its exhibit ID in the report.

Question 30. Identify errors and alternative explanations.

A snapshot may predate or postdate the disputed event; its creation timestamp is not the file’s creation timestamp. Timezone errors, clock drift, NTFS timestamp behavior, restore operations, file synchronization, antivirus quarantine, automatic cleanup and another user’s actions may explain missing or changed files.

A deleted file might be moved, renamed or removed by policy. Carved fragments and shadow-copy versions may be incomplete. State which explanations the evidence excludes and which remain unresolved.

MULTI-SOURCE CYBER INCIDENT INVESTIGATION

UNIQUE CASE SCENARIO

Scenario: The student portal account takeover. A university online-service account is used from an unfamiliar session; investigators examine a student laptop, browser history, account email, mobile device and provider logs to distinguish legitimate access from credential misuse.

SCOPE AND ASSUMPTIONS

The identifiers, times, locations and log entries in this example are fictional. Examination of private messages, mobile devices or provider-held records must be properly authorized.

Question 31. Identify evidence from Windows and other operating systems.

Windows sources include event logs, registry hives, user profiles, browser databases, link files, jump lists, Prefetch where enabled, USB records, scheduled tasks, DNS/cache artifacts and file metadata.

On macOS, relevant sources can include Unified Logs, FSEvents, Spotlight metadata and user Library application data; on Linux, systemd journal/syslog, authentication logs, shell histories and home-directory artifacts may be useful. Capture OS version, clock and encryption state, and validate each artifact’s actual availability rather than assuming it exists.

Question 32. Explain browser artifacts and interpretation.

Browser history, cache, cookies, downloads, saved-session data, autofill, extensions and local storage can reveal visited URLs, file interactions and authenticated sessions.

However, private browsing, clearing history, cross-device sync, background requests, prefetching and shared profiles can create missing or misleading traces. A URL visit does not necessarily mean a user intentionally viewed or submitted a page; correlate browser evidence with server logs and account activity, and avoid exposing unnecessary private browsing data.

Question 33. Identify email and messaging evidence and correlations.

Potential artifacts include message headers, sender/recipient addresses, message IDs, timestamps, authentication results, attachments and hashes, links, delivery status, chat exports and provider audit logs.

Correlate a password-reset email with an account reset event, an attachment hash with a downloaded file, or a message URL with browser activity. Distinguish server-received times from

device display times; documented observations can be altered and forwarded messages do not establish original authorship without supporting metadata.

Question 34. Discuss mobile forensic considerations and challenges.

Record make/model, OS version, lock/encryption state, SIM/eSIM identifiers and network connectivity. Preserve the device's current state according to an approved mobile-acquisition plan; airplane mode or shutdown may affect accessibility and volatile evidence, so document decisions.

Acquire logical or supported file-system data with validated tools and hash exports where feasible. Challenges include strong encryption, app sandboxing, cloud-only data, disappearing messages, device locks, rapid OS changes and the risk of remote wipe. Avoid unauthorized bypasses or invasive collection beyond scope.

Question 35. Discuss Internet and cloud evidence.

Possible sources of evidence include identity-provider sign-in logs, MFA events, IP addresses, user agents, token/session issuance and revocation, cloud object access/share logs, DNS and proxy logs, service-provider timestamps and account recovery records. Preserve provider records promptly because retention may be short.

Request them through authorized administrative or legal processes and document source, export format, timezone and authenticity. IP geolocation is approximate, NAT/VPNs obscure origins, and an authenticated session may result from token theft rather than direct password entry.

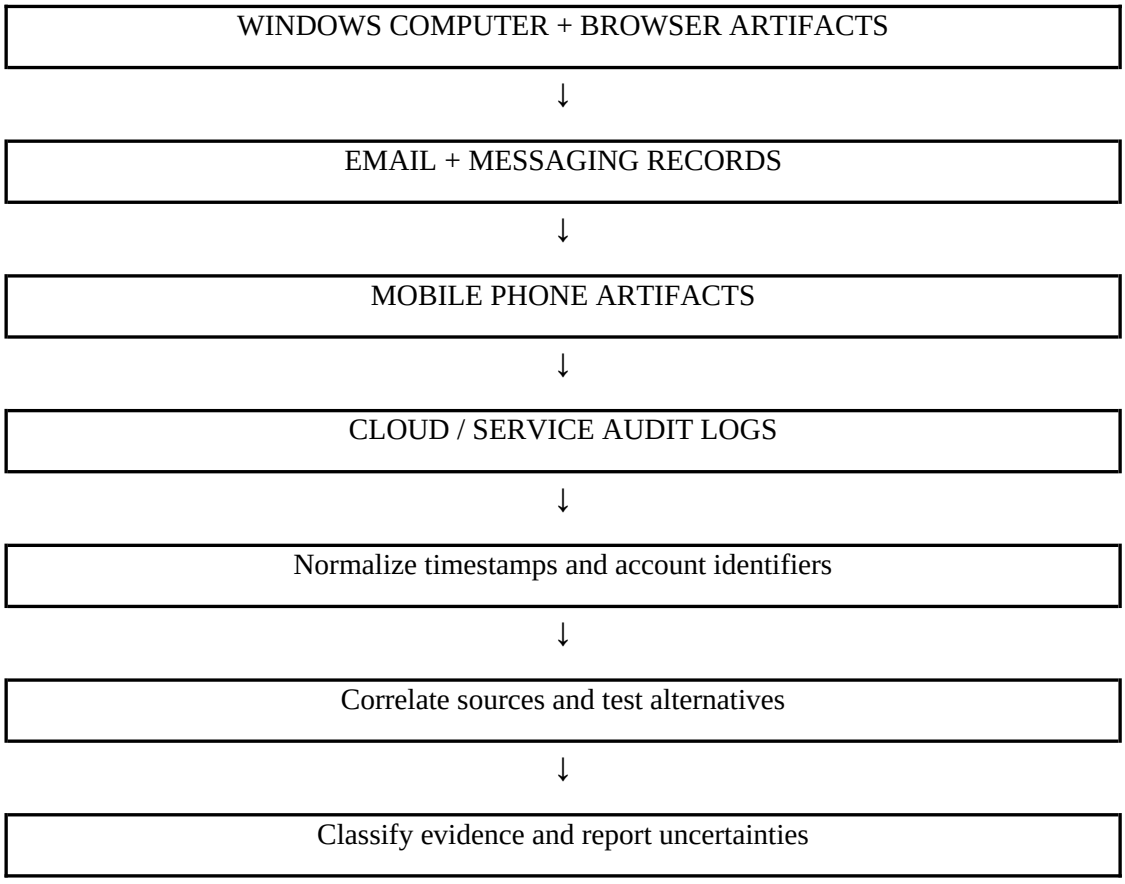
Question 36. Prepare a cross-source correlation table.

The table models a possible investigation; no entry is a real finding.

Source	Hypothetical artifact	UTC time	Correlation / caution
Windows computer	Student profile logon	08:00	Establishes local session, not portal access
Browser	Portal login page visited	08:05	May reflect background or intentional request

Email	MFA notification received	08:06	Correlate provider message ID and headers
Mobile	Push approval event	08:07	Approval may be accidental or coerced
Cloud / identity provider	New session token issued	08:07	Server-side confirmation of account session
Messaging	Suspicious login link shared	07:55	Potential phishing lead; verify sender and URL
Cloud service	Restricted resource accessed	08:10	Confirms service action; not physical operator

MULTI-SOURCE EVIDENCE MAP



Question 37. Explain at least six major multi-source challenges.

1. Different timezones, clock offsets and timestamp precision complicate chronology.

2. Encryption and locked devices limit collection.

3. Cloud providers retain different logs for different periods.

4. Cross-border legal jurisdiction and privacy restrictions can delay lawful access.

5. Shared accounts, NAT, VPNs and stolen tokens weaken identity attribution.

6. Application updates and proprietary formats alter artifact interpretation.

7. Evidence volumes, duplicate synced files and inconsistent identifiers complicate correlation.

8. Remote deletion, ephemeral messages and auto-sync can change evidence after the event. Mitigate these risks through early preservation requests, documented normalization, validated parsers, least-privilege access, hashing and independent corroboration.

Question 38. Prepare a final conclusion distinguishing evidence categories.

Direct evidence of service activity would be an authenticated provider audit entry showing that the account accessed the protected resource at a specified time. Corroborating evidence could include a matching browser session, MFA notification, mobile approval event and corresponding workstation logon.

Unresolved questions include whether the student personally controlled the session, whether credentials or tokens were stolen, whether a phishing message was involved, and whether cloud logs are complete. A defensible provisional conclusion is that the account may have been used for unauthorized access, while attribution to a particular individual remains unproven until independently corroborated.

SUMMARY OF THE INVESTIGATION

Recommended actions: preserve identity-provider logs and relevant device images, compare session identifiers and MFA audit details, investigate phishing indicators within authorized scope, and present separate findings for account activity, device activity and human attribution. Do not equate an account identifier with the identity of the operator.